

1 · RECOMMENDATION

*Meet Mrs. D, 71. Discharged home on a Friday with a new insulin dose.
Eighteen days later she is back through the ED — her third admission this year.*

Illustrative composite patient. Her profile is the segment on the next slide.

Don't build AI yet. Flag the frequent returners with one rule, and fix their discharge.

22.0%

30-day readmission, 2+ prior stays

11.4%

everyone else's baseline

2 · EVIDENCE

Mrs. D isn't rare: risk follows prior admissions, not diagnosis

Framing: one row = one hospital stay. Outcome = readmitted within 30 days, the window payers penalize and discharge care can change.

Prior-year stays	Encounters	30-day rate	Share of readmits
0	66.7%	8.6%	50.3%
1	19.1%	13.3%	22.2%
2	7.3%	17.9%	11.6%
3 or more	6.9%	26.4%	15.9%

Segment (2+ stays, shaded): 14.2% of encounters, **27.5% of readmissions**. By primary diagnosis, every group sits between 9.5% and 13.1%.

27.5%
when the primary diagnosis is diabetes itself

34.7%
discharged to rehab (small group, n=262)

12% vs 17%
A1C tested in the segment vs overall: a process gap

3 · BUSINESS VALUE AND BREAK-EVEN

Targeting halves the bar the intervention must clear

Required relative reduction =
cost per patient ÷ (readmission rate × value of one avoided readmission)

Per 1,000 discharges	Segment only	Everyone
Expected readmits	220	114
Program cost	\$400k	\$400k
Avoided to break even	33	33
Relative cut needed	~15%	~29%

Range if one avoided readmission is worth \$8k–\$15k: the segment needs a 12%–23% cut.

Who captures it?

Under fee-for-service, an avoided readmission is lost revenue for the hospital.

Value reaches the hospital only through penalties avoided, value-based or capitated contracts, or freed bed capacity.

The data holds no cost, revenue or payment rules: CFO to confirm.

ASSUMPTIONS, not in data: \$400 per patient (follow-up + med reconciliation), \$12,000 per avoided readmission · rates are FACTS from the dataset

4 · INTERVENTION

A one-line rule gets most of what a model would

Process ✓

Rules ✓

Analytics

ML

GenAI

Agentic AI

Flag top 14%	Rule: 2+ stays	Boosted model
Precision	21.7%	25.2%
Readmits caught	26.6%	30.8%
AUC	0.60	0.67

Model gain $\approx$ 35 more readmits caught per 1,000 flagged, on 2008 data it would need retraining for.

What Mrs. D gets

- Flagged at discharge: 2+ stays in a year
- Follow-up visit within 7 days
- Medication reconciliation call
- A1C checked before she leaves

5 · UNKNOWNNS AND VALIDATION PLAN

Validate two things in 90 days before anything gets built

Type	Claim	How we close it
Fact	2+ prior stays: 22.0% rate vs 11.4%	Reproducible script
Assumption	\$12k value per avoided readmit, \$400 cost	CFO: current contract mix
Assumption	2008 pattern holds in our hospitals	Rerun rule on 12 months local data
Hypothesis	7-day follow-up cuts segment readmits $\geq 15\%$	90-day pilot vs matched control

Go / no-go at day 90

Scale if the cut clears break-even under our real contracts.
Revisit ML only if rule precision becomes the bottleneck.

*For Mrs. D, the fix is a clinic slot and a phone call,
not a model.*